

CAN 304

Computer Systems Security

Tutorial 6. Cryptography in Blockchain and Web3

Week 6: 2026-04-09, 14:00-15:00, Thursday

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Keynote at ICOW3 2025
Seoul, Korea
November 4-5, 2025



ICoW3 is a conference of the Association for Korea Society of Blockchain (KSB).

Web3.0:

Data Sovereignty as the Key to Unlocking Data Value

An interpretation of Web3.0 from the perspective of data.

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- ❑ Data Sovereignty
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Part 1. Meet Web3.0

Interpreting Web3.0 from the lens of data

The Twin Engines of the Digital Revolution

Blockchain

- ❖ A distributed/decentralized and immutable digital ledger
- ❖ Transforms production relations
- ❖ Provides solution for data ownership
- ❖ The fuel: trust, or rather, mathematics

AI & ML

- ❖ Motivates data ownership
- ❖ ML learns patterns from data to make predictions and decisions
- ❖ Transforms productive force (tools + labor)
- ❖ The fuel: data

Who should own the data and value created from it?



The Price of “Free” in Web2.0

Reviewing Web2.0:

- ❖ Interconnectivity
- ❖ User-Generated Content
- ❖ The Platform Economy

We get amazing services... for “free”.

The real price we pay:

- ❖ Our data is used by platforms for free
- ❖ Value monopolies
- ❖ Worse: losing privacy

We trade privacy for convenience.

- ☐ By checking this box, you agree to our [Terms and Conditions](#)*
- ☐ Check this box if you'd like to receive news, special offers and updates by email.

SAVE



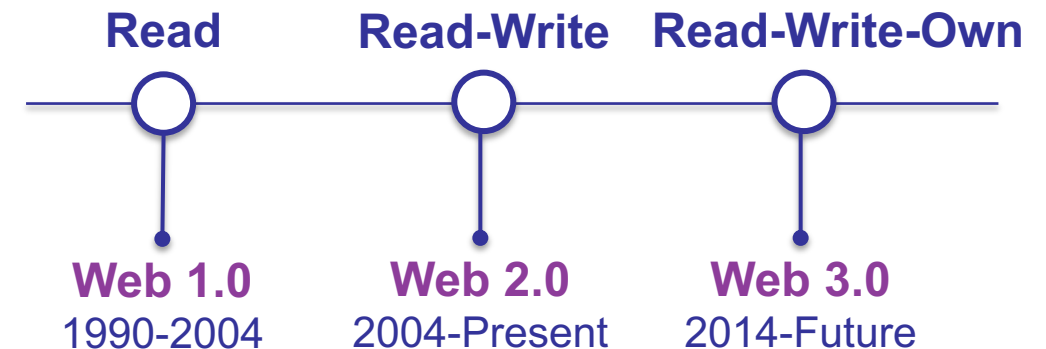
Web3.0: An Internet Revolution About “Ownership”

Web3.0/Web3: The next generation of the internet.

- ❖ Decentralized
- ❖ User-sovereign
- ❖ Built on blockchain technology

Core Comparison:

- ❖ Web1.0: Read (Consume Information)
- ❖ Web2.0: Read + Write (Interact & Create)
- ❖ Web3.0: Read + Write + **Own** (Interact, Create & Own)



Web3.0 or Web3

The term “Web3.0”:

- ❖ Original Meaning (The Semantic Web): Coined by Tim Berners-Lee (the inventor of the web), it envisioned a web where data has defined meaning (semantics), allowing machines to understand and process it intelligently.
- ❖ Modern Meaning: The next generation of the internet

Web3

Stresses decentralization and data ownership.



What Web3.0 is Not

Breaking the Myths:

Web3.0 $\neq$ Crypto Speculation $\neq$ Unlimited Freedom

Myth 1: Web3.0 is just about trading crypto?

→ **Reality:** Cryptocurrency is its incentive layer, not the entirety of the vision.

Myth2: Web3.0 means unlimited freedom and no rules?

→ **Reality:** It's not a lawless space. It's a system governed by transparent code and community consensus, offering new forms of digital governance.



Reassessing the Ownership of Data and Its Value

Technological Revolution

Blockchain and AI advancements are transforming our perception of data

Digital Gold

Data has become the most valuable resource in the digital economy

Ownership Challenges

Traditional internet models face issues with data ownership and value distribution

Web3.0 Evolution

The next stage of internet development will reshape our relationship with data and its value

 **Focus:** This presentation explores Web3.0 through **the lens of data**, examining how data sovereignty unlocks the true value of data in the emerging digital landscape.



Part 2. Data Sovereignty

The key to unlocking data value

Data Sovereignty

What is Data Sovereignty?

It is the collection of your **control**, **access**, and **portability rights** over your personal data.

Why is it the “Key”?

Without sovereignty, you cannot authorize, you cannot trade, and you cannot prove the value belongs to you. Sovereignty is the foundation for all value-generating activities.

Analogy:

It's like the property deed to your house. Without it, your house is just a place you live, not an asset you own.



The Tech Stack

Blockchain:

Provides an immutable, public record of ownership (The Ledger).

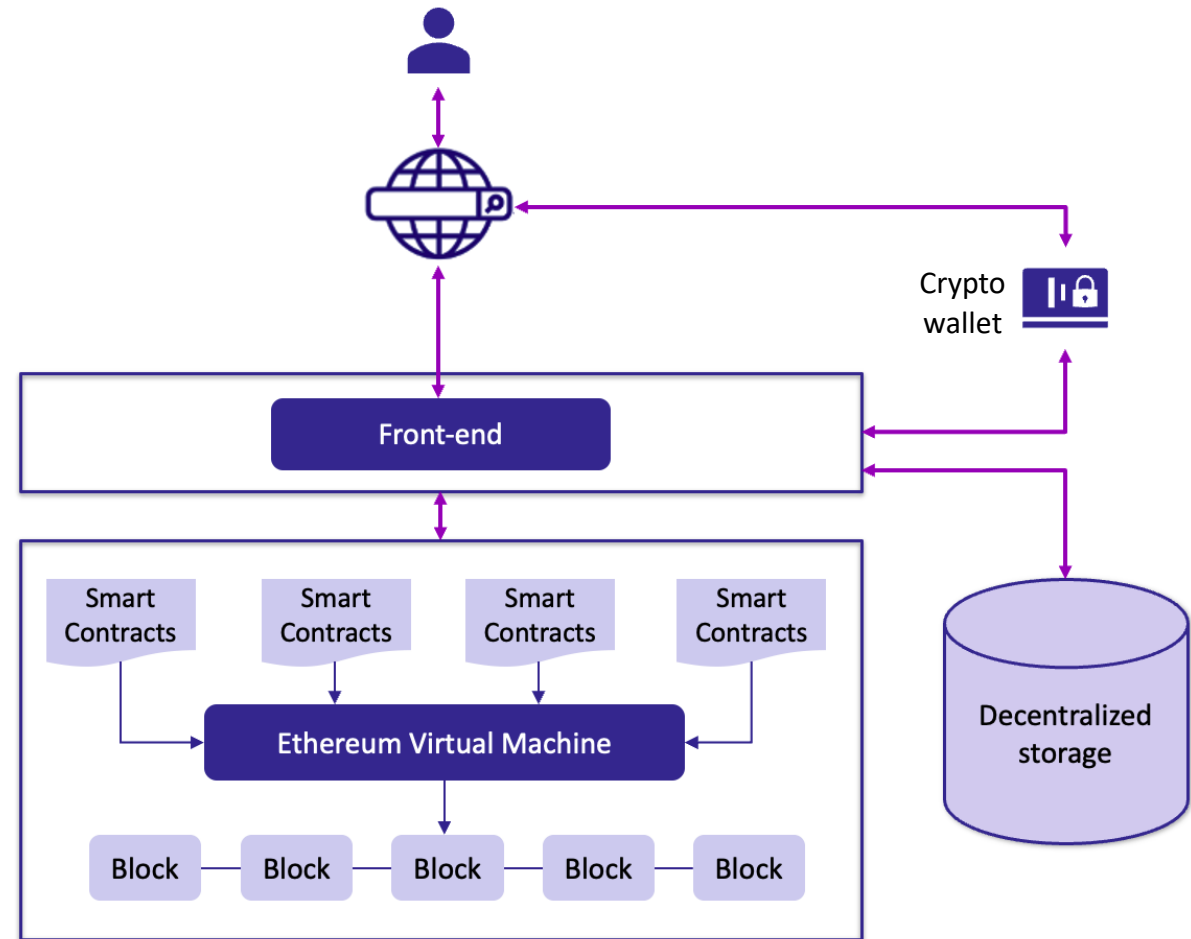
Decentralized Identity (DID):

Provides an independent, verifiable digital identity (The Passport).

Smart Contracts:

Provide self-executing rules for authorization and transactions (The Contract).

Users participant Web3.0 via **Crypto Wallet**



Crypto Wallet

It's Not a Wallet, It's Your **Identity Container**.

Foundation: Digital Signature

DS = (KeyGen, Sign, Verify)

- $\text{KeyGen} \rightarrow (sk, pk)$
- $\text{Sign}(m, sk) \rightarrow \sigma$
- $\text{Verify}(m, \sigma, pk) \rightarrow b \in \{0,1\}$

The **private key** is crucial to ensure ownership of users' data and digital assets in the Web3.0

Three core functions

- ❖ Authentication (proof of origin)
- ❖ Integrity (Proof of integrity)
- ❖ Non-Repudiation (Proof of commitment)



Crypto Wallet Challenges

Example private key in hexadecimal format (for the secp256k1 curve used by Bitcoin/Ethereum):



```
e3b0c44298fc1c149afbf4c8996fb92427ae41e4649b934ca495991b7852b855
```

The Usability Cliff

“Be your own bank” is hard.

Managing long strings of keys and understanding gas fees is a huge barrier for the average user.

The Security Nightmare

Private keys are a single point of failure.

The Recovery Abyss

Forget your key? Lose your device? Your assets are gone forever.



KeyShield: Protecting Your Private Key

Method:

Secret Sharing

Shamir's (t, n) -threshold secret sharing

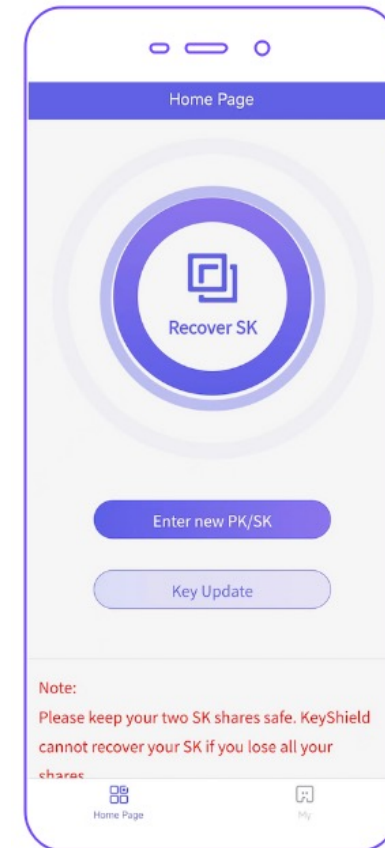
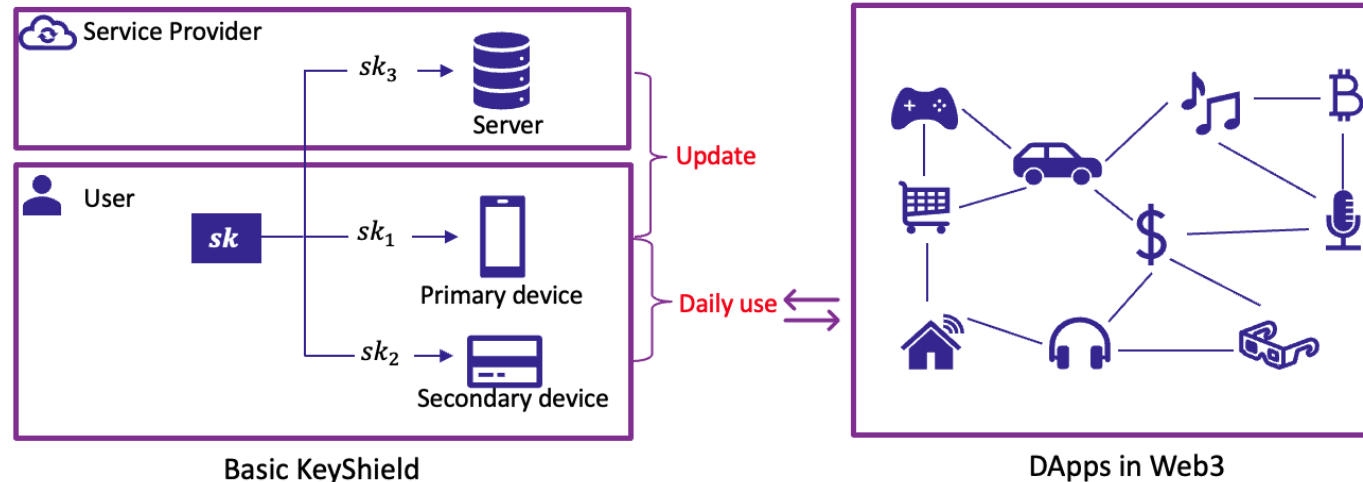
$$f(x) = a_0 + a_1x + \cdots + a_{t-1}x^{t-1}$$

Distribution: $(x_1, y_1), \dots, (x_n, y_n)$

Reconstruction: $a_0 = \sum_1^t y_i \prod_{j=1, x_j \neq x_i}^t \frac{x_i}{x_i - x_j}$

Functionality

- ❖ Daily use
- ❖ Recover and update

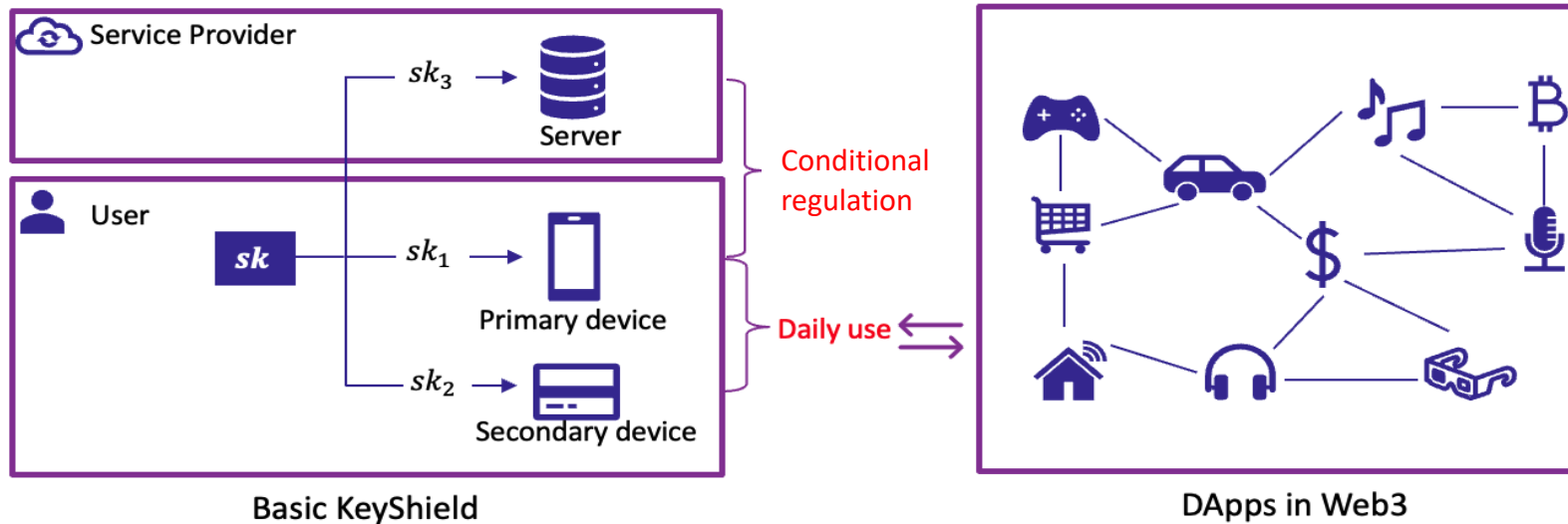


KeyShield Can Do More

Addressing conditionally regulatable Web3

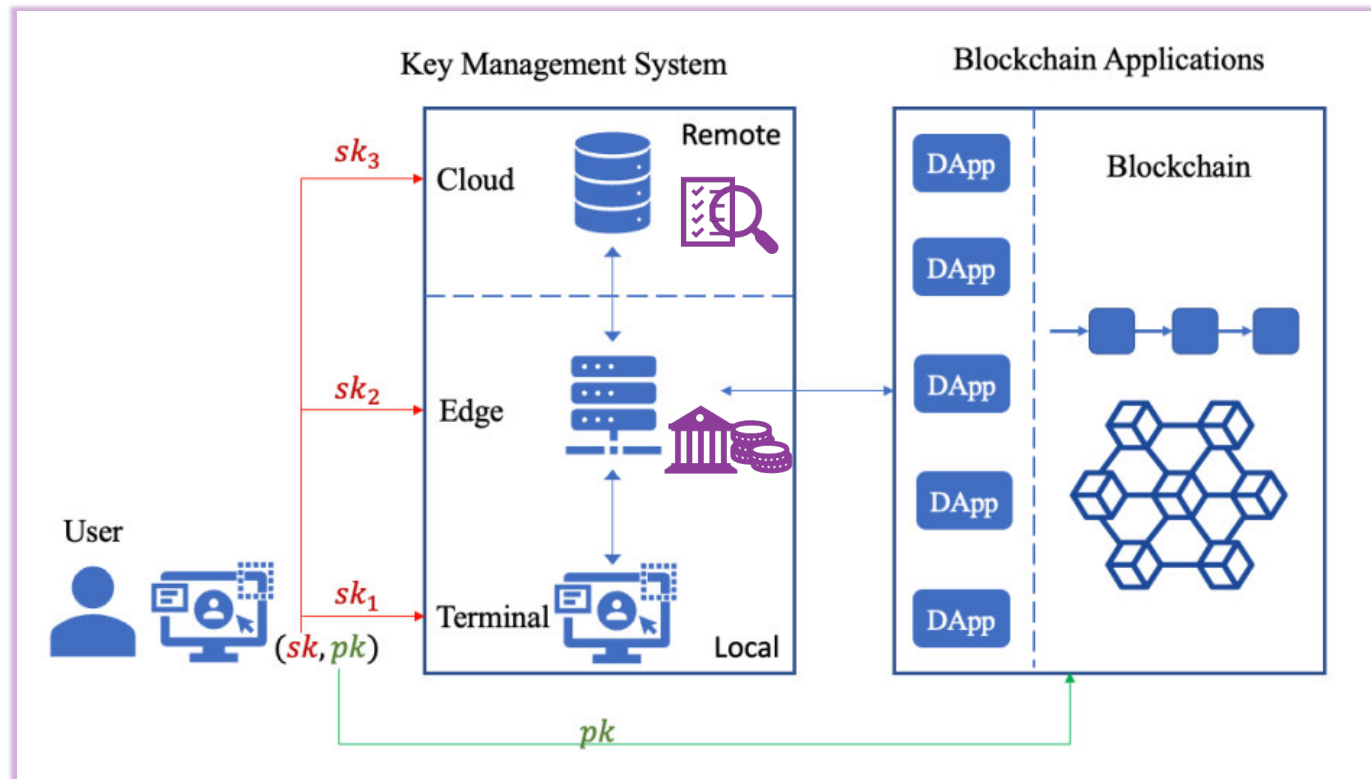
Web3.0 is not a lawless space.

It's a system governed by transparent code and community consensus, offering new forms of digital governance.



KeyShield Can Do More

Cloud-edge-end based key management for DApps



Zhang, J., & Zhang, F. (2024). Cloud-edge-end based key management for decentralized applications. *Journal of Surveillance, Security and Safety*, 5(4), 213-233. 17 Dec 2024



Updatable & Revocable Signatures

Revocable Signature

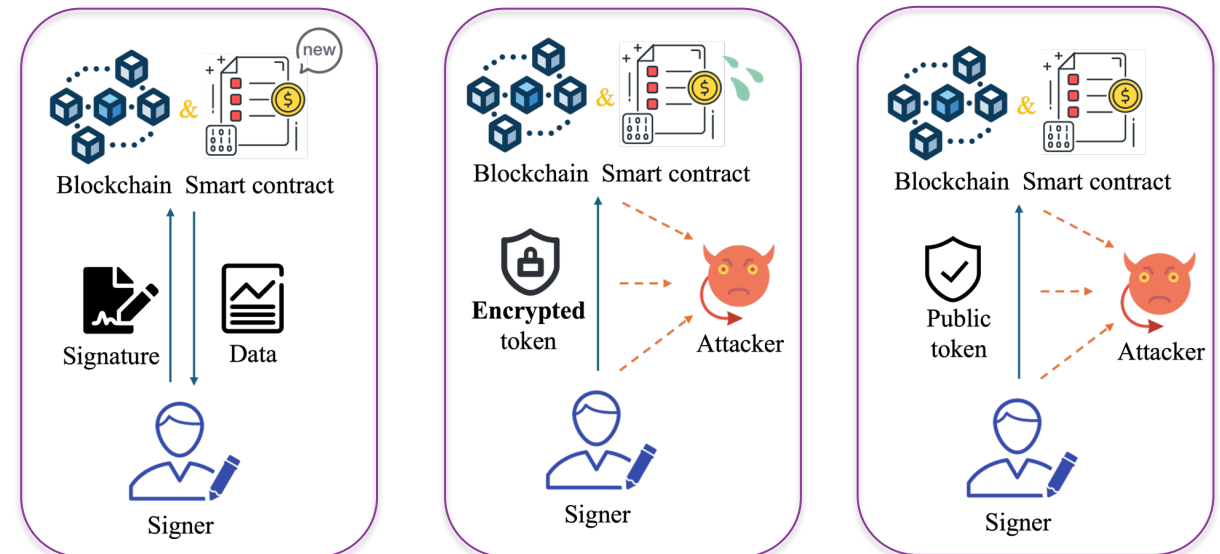
- ❖ Prove that a valid signature issued by private key thief is not authenticated.
- ❖ Revoke valid but unauthorized signatures!

$$\begin{aligned} \text{DS} &= (\text{KeyGen}, \text{Sign}, \text{Verify}) \\ \text{KeyGen} &\rightarrow (sk, pk) \\ \text{Sign}(m, sk) &\rightarrow \sigma \\ \text{Verify}(m, \sigma, pk) &\rightarrow b \in \{0, 1\} \end{aligned}$$

Xia, Y., **Zhang, J.**, **Man, K. L.**, etc. When Valid Isn't Volitional: Revocable Signature with Publicly Verifiable Exposure of Valid but Unauthorized Signatures. *Cybersecurity*, 9 (97), 2026.

Updatable Signature

- ❖ Allows updating signatures so that they can be verified using a new key.
- ❖ The signer rotate its key pairs and signatures by issuing an update token.



Yin, H., **Zhang, J.**, Li, W., Dong, Y., Lim, E. G., & Wojtczak, D. (2025). Updatable Signature with public tokens. *Journal of Information Security and Applications*, 91, Article 104058. June 2025

Part 3. Trusted Data Flow

The ultimate goal: from “static asset” to “flowing value”

Trusted Data Flow

What is Trusted Data Flow?

The secure movement of data between **different parties** under **conditions** where its source is trustworthy, its use is controllable, and its process is transparent, thereby generating value.

Why is it crucial?

Merely having sovereignty (owning the key) is not enough.

Creator of data $\neq$ Exploiter of data value

Let data participate in economic activities to **realize its value**. Flow is the process of value realization.



Data Trading

Data trading: The peer-to-peer exchange of data assets for value, without relying on a central intermediary.

Four basic stages:

- ❖ Data listing
- ❖ Data discovery phase
- ❖ Matching stage
- ❖ Execution phase

Challenges:

- ❖ Data quality and compliance
- ❖ Pricing
- ❖ Privacy
- ❖ Behavior fairness



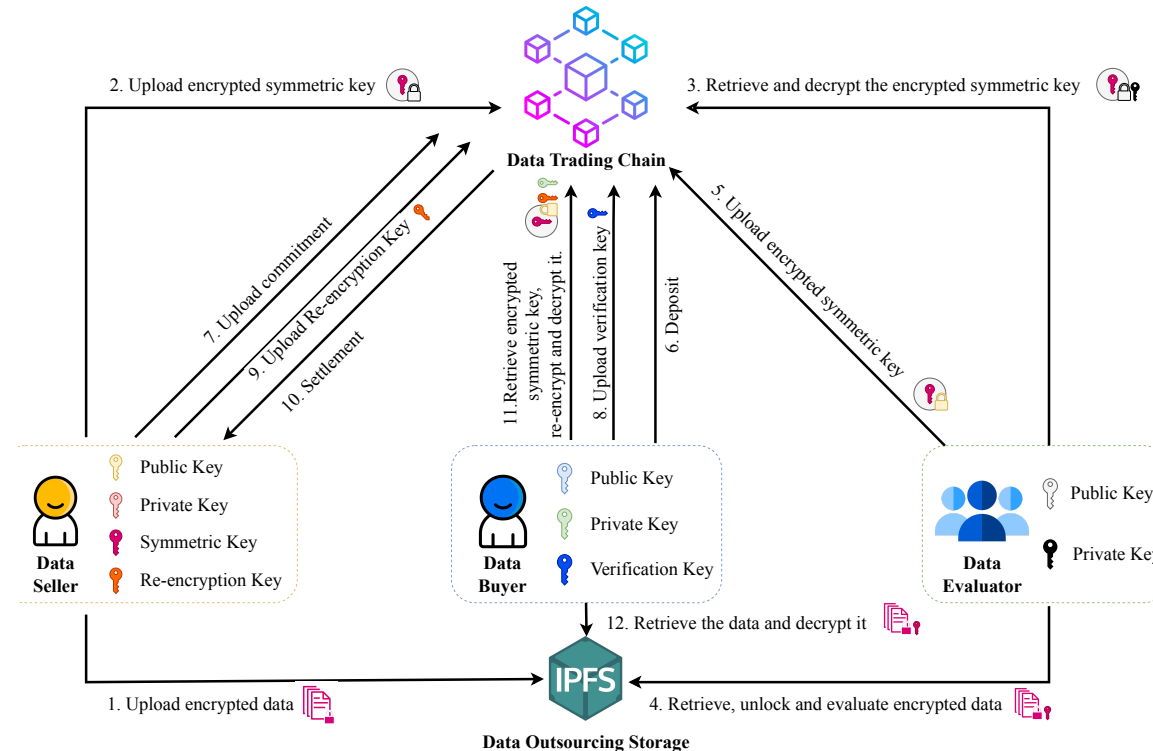
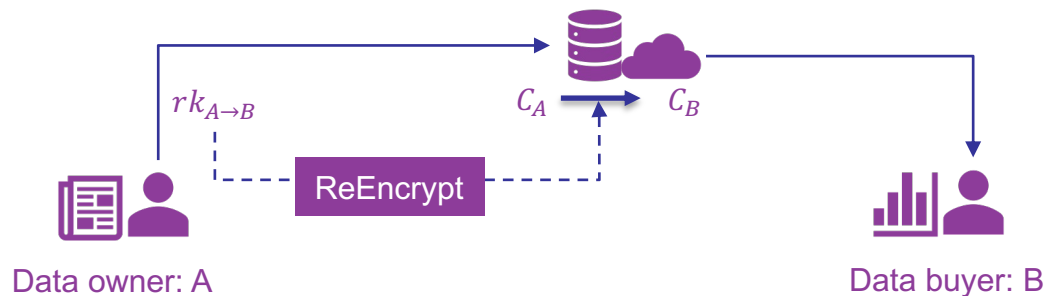
Fair & Secure Data Trading

Dishonest behaviors violating fairness:

- ❖ Data seller may provide wrong or incomplete data after receiving money from a buyer.
- ❖ Data buyer may refuse to pay the requested money after receiving the correct data from a seller.

Method: Proxy Re-Encryption (PRE)

Allows a third party (a proxy) to transform ciphertext from being decryptable by one person to being decryptable by another, without ever seeing the underlying plaintext or the private keys.



Ji, Z., Zhang, J., Man, K. L., Guan, S., & Huynh, N. V.. Fair Data Trading on Blockchain Through Verifiable Proxy Re-Encryption. *IEEE Transactions on Computational Social Systems*, 13 (1), 2026

Fair & Secure Data Trading

Privacy Issue & Confidential Trading

❖ Data seller:

Data content should remain confidential and only be accessed by authorized parties.

❖ Data buyer:

Searching/purchasing interest: sensitive business strategies leakage to competitors

Evaluation criteria: seller may manipulate or fabricate data to meet them

Methods:

❖ Private Information Retrieval

❖ Two-party Homomorphic Encryption



The Ultimate Goal:

How can we use sensitive data for AI training without ever seeing the raw data itself?

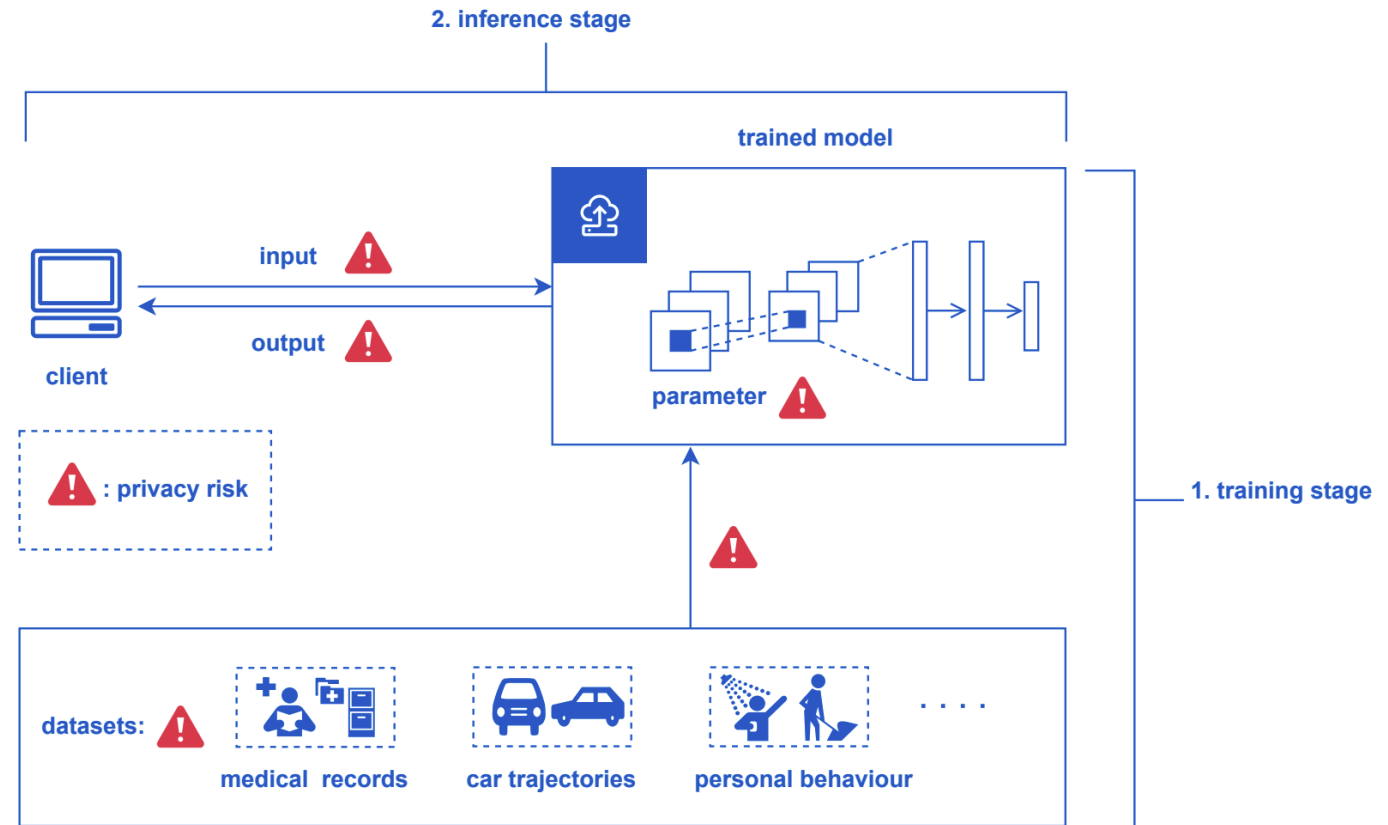
Privacy-Preserved ML

Training stage

- ❖ Datasets

Inference stage

- ❖ Model parameters
- ❖ User input and inference results



Privacy-Preserved ML - Training

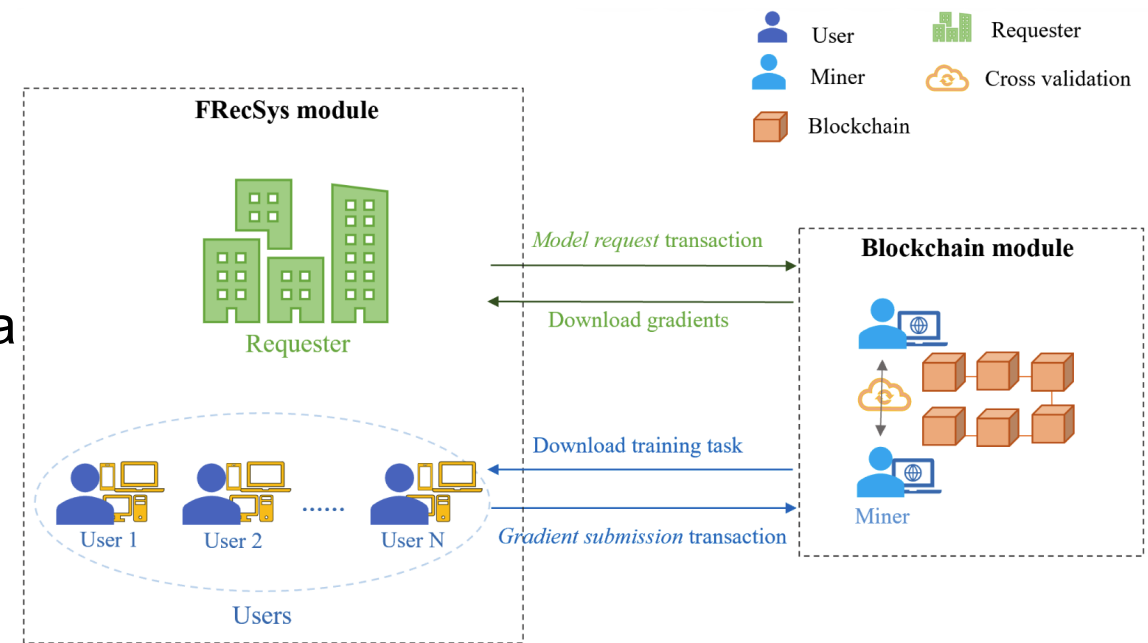
Centralized Learning

To train an AI, must pool all sensitive data (e.g., patient records) in one central server.
This creates a massive privacy and security risk.

Federated Learning

Keep the data where it is. Instead, send the AI model to the data.

- ❖ Distribute: The global model is sent to each data source (e.g., Hospital A, Hospital B).
- ❖ Train Locally: Each hospital trains the model on its own private, local data.
- ❖ Aggregate: Only the learned insights (model updates, not the patient data) are sent back and combined to improve the global model.



Hou, D., **Zhang, J.** Bfrecsys: A blockchain-based federated matrix factorization for recommendation systems. *IEEE International Conference on Big Data*, 15-18 December 2023

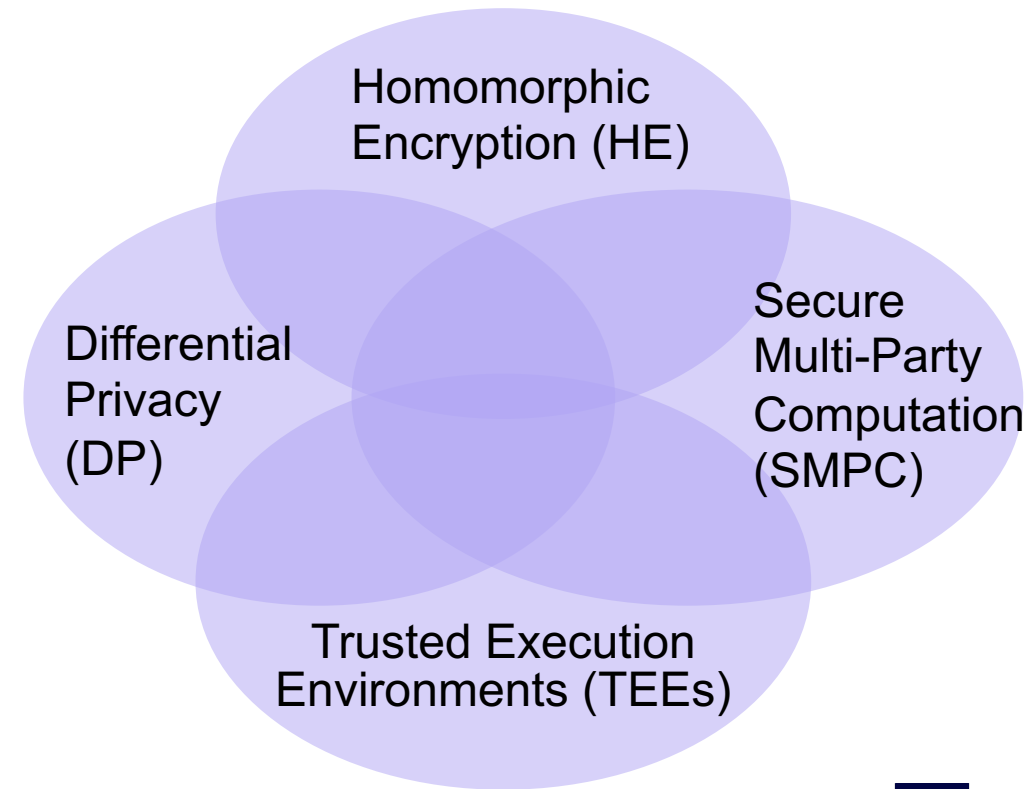
Privacy-Preserved ML - Inference

User's Privacy

- ❖ The Risk: The model owner could see the user's sensitive data (e.g., symptoms, financial info).
- ❖ The Goal: Ensure the model owner learns nothing about the user's queries and inference results.

Model's Privacy

- ❖ The Risk: The user could steal, copy, or reverse-engineer the valuable model that the company spent millions to train.
- ❖ The Goal: Ensure the user learns nothing about the model's internal parameters (weights & biases).



Part 4. Conclusion

Core Insights & Future Outlook

Insights

- ❖ Data sovereignty – who owns data
- ❖ Trusted data flow – who owns the value created from data
- ❖ Privacy-preserved ML – creating value from data

Future Outlook

Data is invaluable. While Web3.0 offers the foundational framework to maximize its utility and worth, we still face challenges. Overcoming them requires a unified and collaborative effort from all of us.





THANK YOU